

Are You Getting What You Pay For? Auditing Model Substitution in LLM APIs

Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation Wins

Trust gap: API users cannot directly verify served model identity in black-box deployments

Paper: [arXiv:2504.04715](https://arxiv.org/abs/2504.04715)

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Economic driver: providers can reduce GPU cost while preserving premium API pricing

Substitution types: smaller model, FP8/INT8 quantized variant, fine-tuned/updated version, different model family

Formal audit framing: $H_0: P_{\text{api}}(y|x) = P_{\text{spec}}(y|x)$ vs $H_1: P_{\text{api}}(y|x) \neq P_{\text{spec}}(y|x)$

Core challenge: only output samples are observable to external auditors

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Higher Temperature Amplifies Performance Gaps but Increases Noise

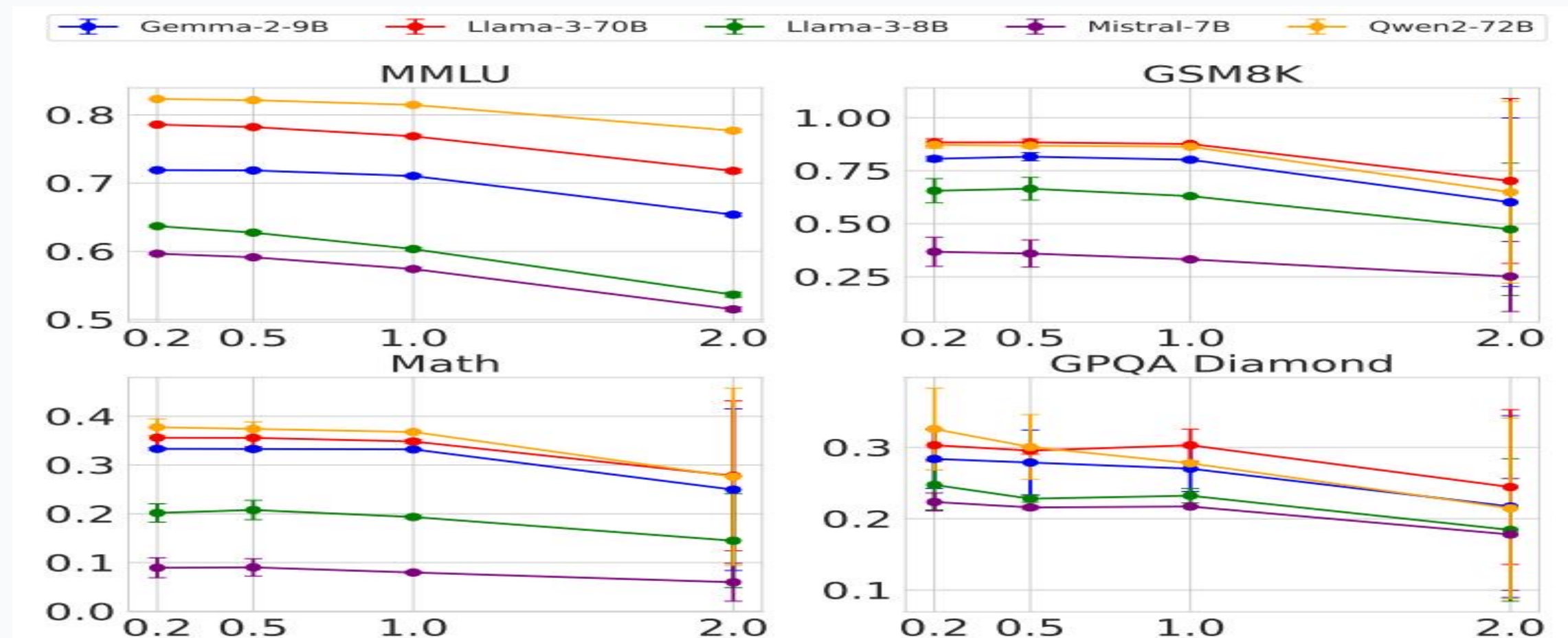


Figure 1: Benchmark accuracy vs temperature (0.2, 0.5, 1.0, 2.0) across five model families.

At high temperature, all models degrade and substitution signal becomes noisier.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure 5: Same Llama-3-8B model yields varying log-probabilities across frameworks and GPUs.
Nondeterminism across transformers/vLLM versions and A100/H100 breaks stable fingerprinting.

Randomized Routing at Low Rates Makes Statistical Detection Query-Proh

Attack: provider routes each query to substitute model with probability p , else to specified model.

Observed distribution: $P_{\text{mix}}(y|x) = (1-p)P_{\text{spec}}(y|x) + pP_{\text{sub}}(y|x)$.

As $p \rightarrow 0$, P_{mix} converges to P_{spec} ; statistical distance shrinks and required sample size explodes.

Implication: low-rate substitution can evade practical black-box audits under realistic query budgets.

Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Weakness	Practical Limitation
Text-based statistical tests (e.g., MMD)	Low sensitivity to subtle substitutions	Requires very large query counts; weak under realistic budgets
Log-probability fingerprinting	Inference-stack nondeterminism changes scores	Cannot separate infrastructure drift from true model substitution
Adversarial countermeasures (benchmark evasion, randomized routing)	Provider can strategically mask substitution behavior	Detection becomes query-prohibitive and operationally fragile

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

TEE attestation binds execution identity to measured code, runtime, and model artifacts.

Provider can prove that expected model weights and inference binary are loaded in confidential hardware.

Security property: cryptographic verification replaces probabilistic inference from outputs.

Deployment path: NVIDIA confidential computing on H100-class infrastructure with modest overhead.

Compared to software-only tests: stronger guarantees, lower audit ambiguity, production viability today.

Key Takeaways: Close the Verification Gap with Hardware Attestation

Model substitution is economically motivated and difficult to detect from outputs alone.

FP8 quantization often stays within benchmark error bars, masking covert substitution.

Inference nondeterminism invalidates stable log-probability fingerprints.

Randomized substitution and benchmark evasion make software audits query-prohibitive.

Call to action: adopt TEE-based attestation as baseline integrity control for commercial LLM APIs.